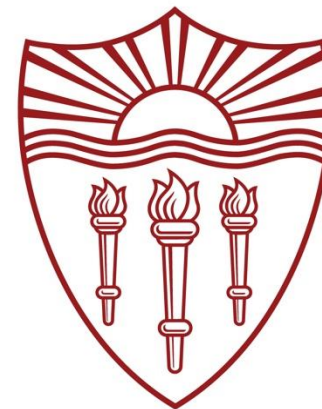




LA's BEST @ USC

Introduction to Biostatistics &
A Motivation for Statistical Learning

Trevor A. Pickering, PhD

Associate Professor

USC Division of Biostatistics and Health Data Science

Overview

- “About me” - AKA - “Why the heck do I like stats?”
- What is biostatistics?
- Types of data
- Thinking statistically
- Describing a sample: distributions, percentiles, central tendency, variation

An FAQ About... Me?

Who am I?

- Assistant Professor of Biostatistics at the Keck School of Medicine of USC

What do I do?

- Short answer: Teaching (40%), Research (40%), "Service" (20%)
- Co-Director of the CTSI BERD Biostatistics Consulting core
- Director of the MS programs in Biostatistics and Public Health Data Science

An FAQ About... Me?

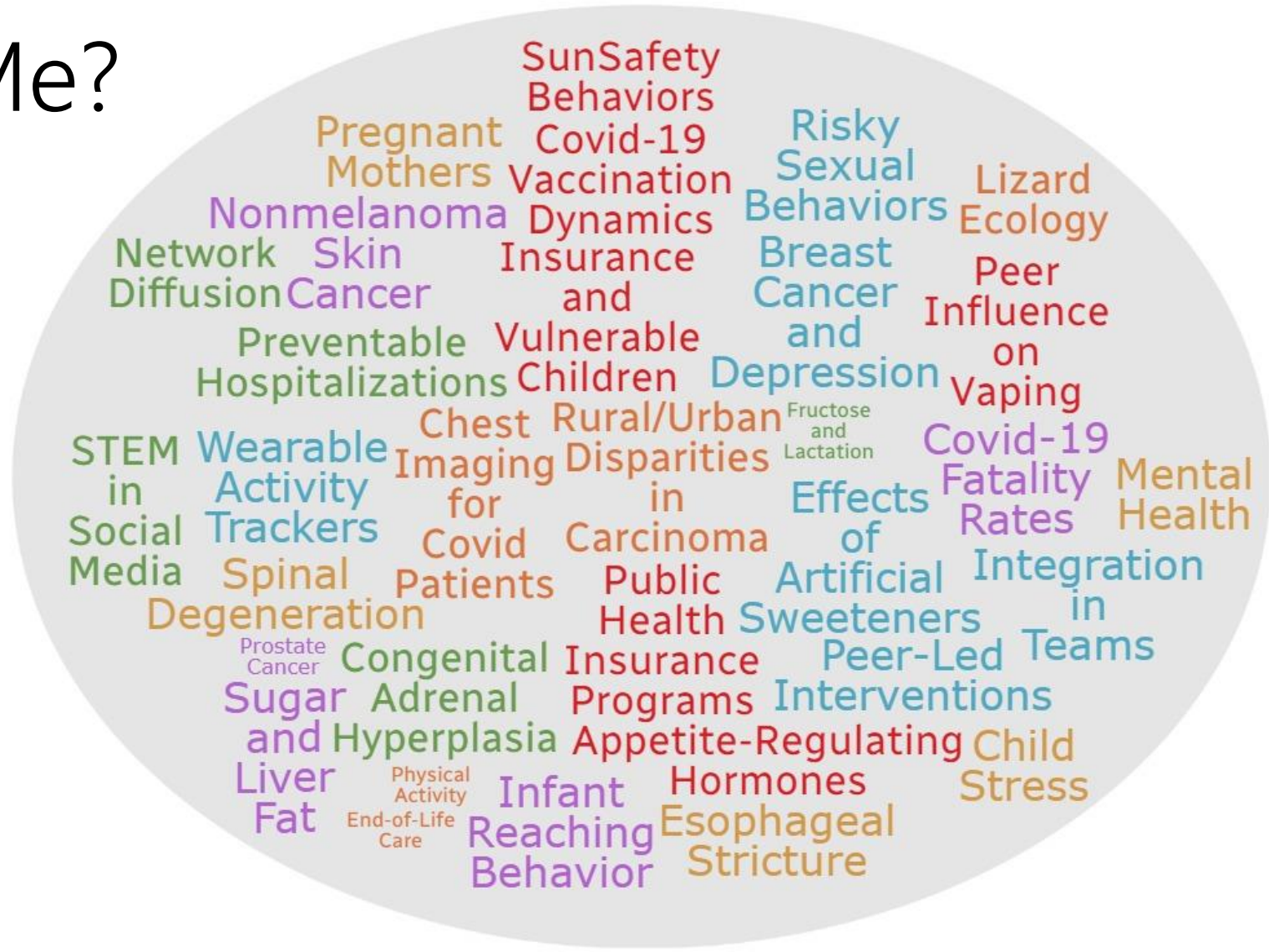
How did I end up here?

- BS in Mathematical Biology studying measurement of lizards (lizard ecology)
- MS in Biostatistics studying air pollution and lung function in adolescents
- PhD in Preventive Medicine - Health Behavior studying peer-led interventions in schools



An FAQ About... Me?

And I enjoy exploring data!

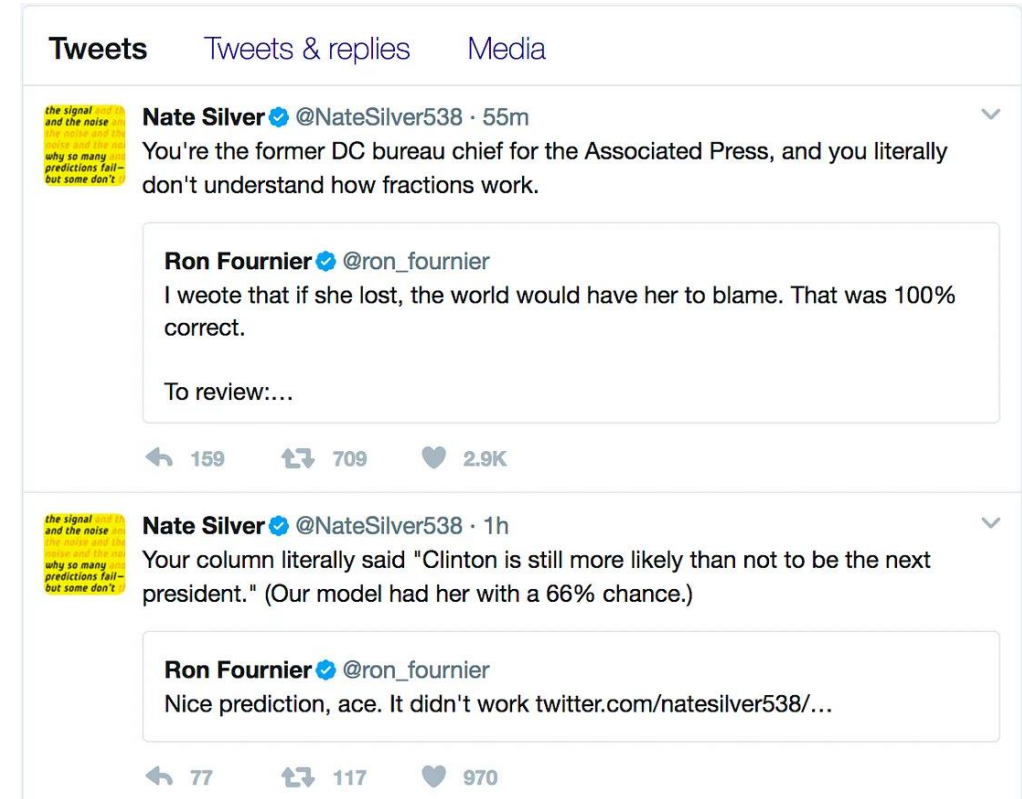


Why do I enjoy statistics?

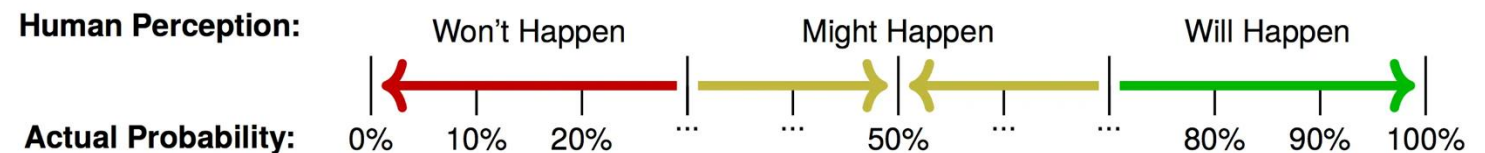
Humans are **BAD** at assessing things!

Example: People frequently misinterpret probability

<https://towardsdatascience.com/humans-are-bad-at-probability-834980e719a3>.



Why Humans are Bad at Thinking About Chance

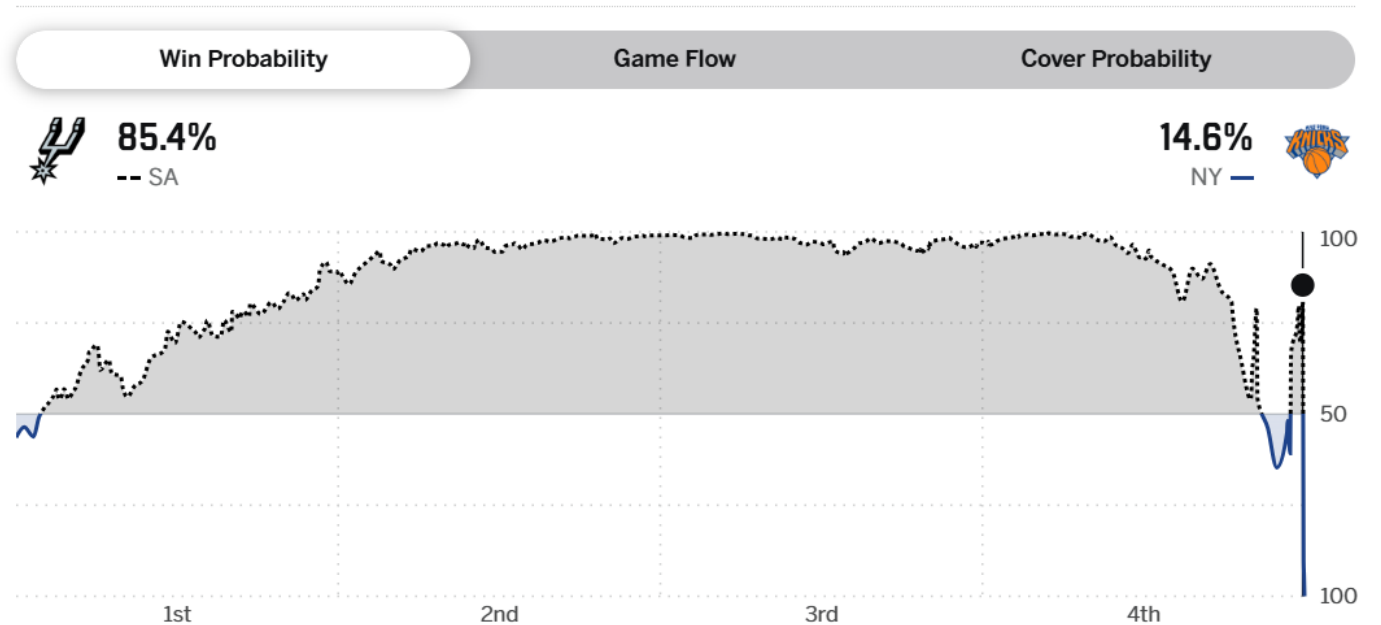


Why do I enjoy statistics?

Take Game 4 of the NBA playoffs as an example

(This is right before Anunoby's tip-in.)

PROBABILITIES & GAME FLOW



Missed 3PT

4.4 - 4th

Jalen Brunson misses 30-foot three point pullup jump shot

106 105

SA NY

Why do I enjoy statistics?

Statistics makes us more informed about our world!

Example: Group Fitness

GROUP FITNESS: THE 3 ELEMENTS THAT Make It Work

DID YOU KNOW THERE ARE 3 KEY FACTORS THAT AFFECT
A NEW PERSON'S GROUP FITNESS EXPERIENCE?

SMARTSTART

The level of satisfaction that our members experience in class will determine whether or not they come back the following week. Consider this: 50% of people who start a new exercise regime will drop out within the first 6 months. If we understand how we can influence our participants' satisfaction levels, this will help us to get new exercisers addicted to fitness.

THE THREE KEY FACTORS

Past research has highlighted three variables that are key to improving satisfaction during exercise classes. These are: intensity, social connection and competence.

Intensity

Previous research has indicated that exercising at high intensity can result in a level of discomfort that can be associated with reduced satisfaction, particularly for beginners.

Social Connection

Feeling connected to the group and the instructor has been found to be important to

THE TEST

To see how these variables affect class satisfaction in LES MILLS™ programs, we completed a 30-week intervention with 25, non-active, but otherwise healthy, adults between the ages of 25 and 40. The participants completed a 6-week familiarization protocol followed by two 12-week blocks of 6 or 7 Les Mills classes per week. After each class, they completed surveys that included questions related to their levels of satisfaction with the class, what they thought of the instructor, the levels of intensity, their connection to the group and the instructor, and their levels of competence.

THE RESULTS

There was a difference between the levels of satisfaction and **Intensity** depending on the type of class. For example, when the participants completed a cardio class such as BODYATTACK™, BODYCOMBAT™, BODYSTEP™, or RPM™ and rated the class as being intense and challenging, they also rated the class as being highly satisfying. On the other hand, if they found a BODYPUMP™ or BODYBALANCE™/BODYFLOW® class to be more intense than anticipated, then they were not as satisfied with the overall experience.

Feeling **Connected** to the group and the instructor was extremely important to the levels of satisfaction in the class, and this was particularly evident during the first 12-week block. This connection allowed participants to cope with the discomfort of exercise, and therefore increased their enjoyment of the workout.

Finally, when we looked at **Competence**, the beginner participants were more satisfied when there was technical and clear instruction, enabling them to execute the movements well. Then, once they felt they had developed the skills required to complete

WHAT DOES YOUR CLASS NEED?

1. Remember more structure than your pushed to but we need slowly in E and BODY
2. Using clear people to in the early levels, giving energy of
3. It's vital to you the For tips or connection the Educators Directors

Don't forget to **template** to provide new after the first

www.lesmills.com

Perceptions of the activity, the social climate, and the self during group exercise classes regulate intrinsic satisfaction

 **Jaclyn P. Maher**^{1,2*},  **Jinger S. Gottschall**¹ and  **David E. Conroy**^{1,3}

¹Department of Kinesiology, Pennsylvania State University, University Park, PA, USA

²Department of Preventive Medicine, University of Southern California, Los Angeles, CA, USA

³Department of Preventive Medicine, Northwestern University, Chicago, IL, USA

Engaging in regular physical activity is a challenging task for many adults. Intrinsic satisfaction with exercise classes is thought to promote adherence to physical activity. This study examined the characteristics of exercise classes that impact within-person changes in intrinsic satisfaction over the course of an extended group exercise program. A 30-week physical activity trial was conducted with assessments at the end of each class. Community-living adults ($n = 29$) were instructed to complete at least six group exercise classes each week and, following each exercise class, complete a questionnaire asking about the characteristics of the class and the participant's evaluation of the class. Intrinsic satisfaction was high, on average, but varied as much within-person from class-to-class as it did between exercisers.

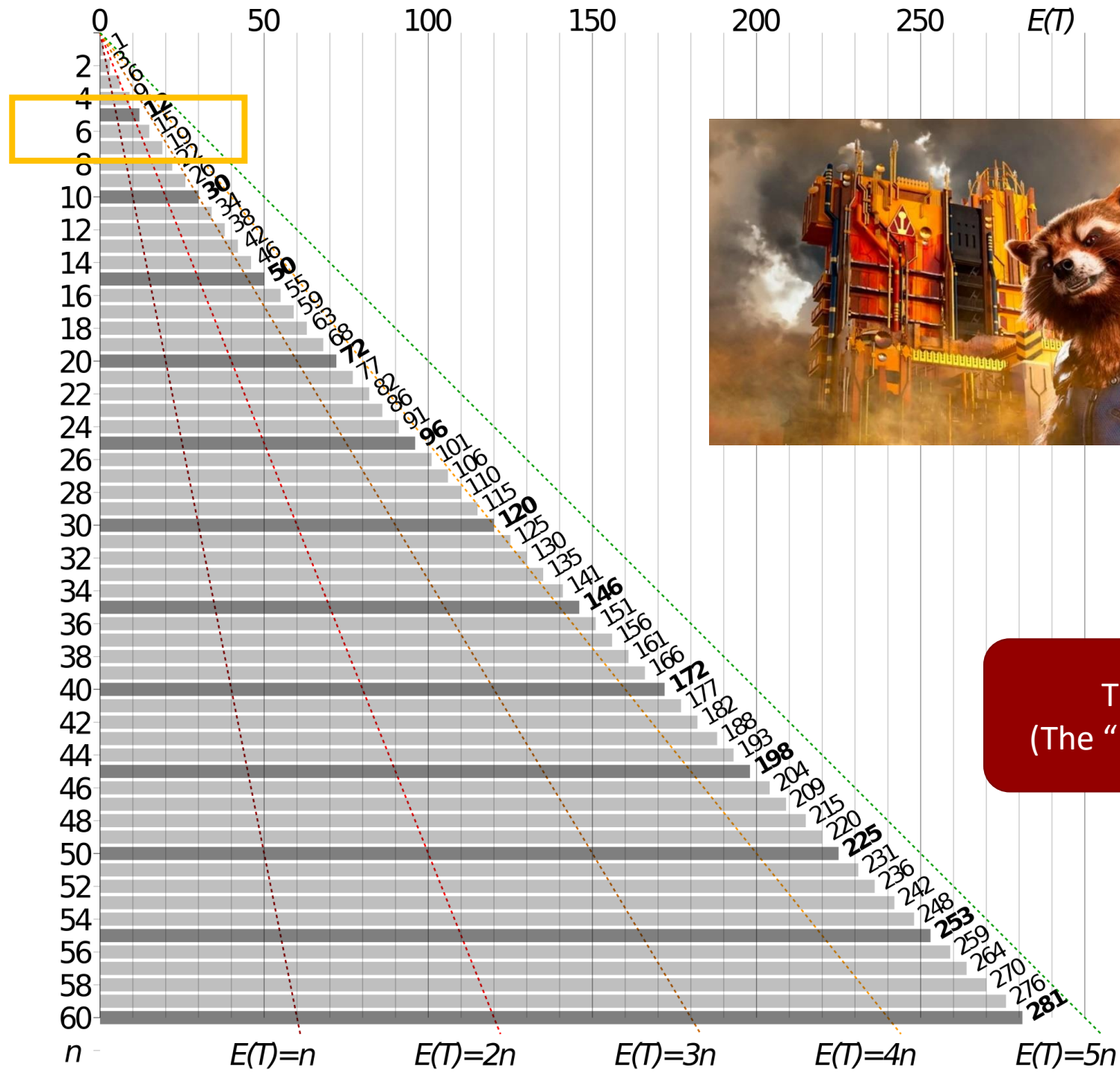
Participants reported the greatest intrinsic satisfaction when classes placed greater emphasis on exercisers' involvement with the group task, feelings of competence, and encouragement from the instructor. For the most part, exercise classes that were more intense than usual were perceived by exercisers as less intrinsically satisfying. Some overall characteristics of the exercise classes were also associated

with intrinsic satisfaction. The social and motivational characteristics of group

Why do I enjoy statistics?

Statistics makes us more informed about our world!

Example: Probabilities of Events



The “coupon collector” problem
(The “Happy Meal toy collector” problem)

Why do I enjoy statistics?

Statistics takes the “I’ve noticed...” or “I think that...” and puts a number to it!

Picture it – London – 1710.

It seems like more males tend to be born than females.

John Arbuthnot examines birth records from 1629 to 1710.

Every year there were more males born than females.

Could this happen due to chance?



Arbuthnot thought this was *divine providence*.
“It is art, not chance, that governs.”

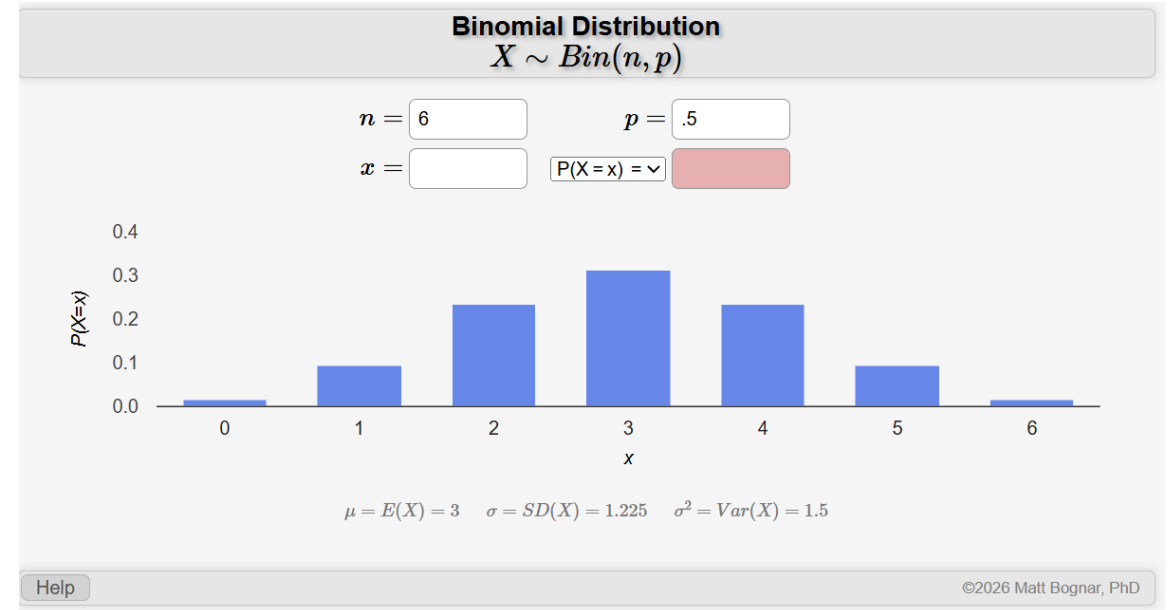
Why do I enjoy statistics?

I will now share psychic predictive powers with the class.

Why do I enjoy statistics?

Some questions to consider:

- How uncommon/improbable would it be for all 6 cards to be guessed correctly?
- Would it be just as surprising if all 6 guesses were incorrect?
- At what number of correct guesses do we stop believing that I'm just randomly guessing the outcome?



X	P
6	0.016
5	0.094
4	0.234
3	0.313

What is biostatistics?

Misconception: it's just calculating things like mean, median, mode

Descriptive Statistics					
Variable	<u>Obs</u>	Mean	<u>Std.Dev.</u>	Min	Max
price	74	6165.257	2949.496	3291	15906
mpg	74	21.297	5.786	12	41
rep78	69	3.406	.99	1	5
headroom	74	2.993	.846	.846	5
trunk	74	13.757	4.277	5	23
weight	74	3019.459	777.194	1760	4840
length	74	187.932	22.266	142	233
turn	74	39.649	4.399	31	51
displacement	74	197.297	91.837	79	425
<u>gear_ratio</u>	74	3.015	.456	2.19	3.89
foreign	74	.297	.46	0	1

What is Biostatistics?

A mathematical body of science dealing with the collection, analysis, interpretation, and presentation of (biological) data.

Many specialized branches

- Statistical genetics
- Environmental statistics
- Clinical trials
- Psychometric statistics

Modern Biostatistics

Really took off in the early 1900s

- Francis Galton – 1907 – examined 787 guesses of the weight of an ox at a county fair. The median was 1208; the actual weight was 1,198. *The Wisdom of the Crowd*.
- Karl Pearson – Early 1900s – *The “standard deviation.” The correlation coefficient.*
- Ronald Fisher – 1920s-30s – *Design of Experiments, Rothamsted Experimental Station*

Modern Biostatistics

Biostatistics in 2026

- The need for biostatisticians and data scientists is expected to increase anywhere from 8% to 36% between now and 2034.
- Some top jobs for individuals in biostatistics are:
 - Data Analyst/Scientist
 - Biostatistician
 - Software Engineer
 - Research Analyst
 - Bioinformaticist

Modern Biostatistics

PERSONAL AND PROFESSIONAL SKILLS OF BIOSTATISTICIANS

Biostatisticians work in offices, in laboratories and in the field conducting a range of tasks, from designing research studies to analyzing and reporting on their results. In addition to technical and statistical skills, biostatisticians require several personal and professional skills.



Written and oral communication



Problem-solving



Critical thinking



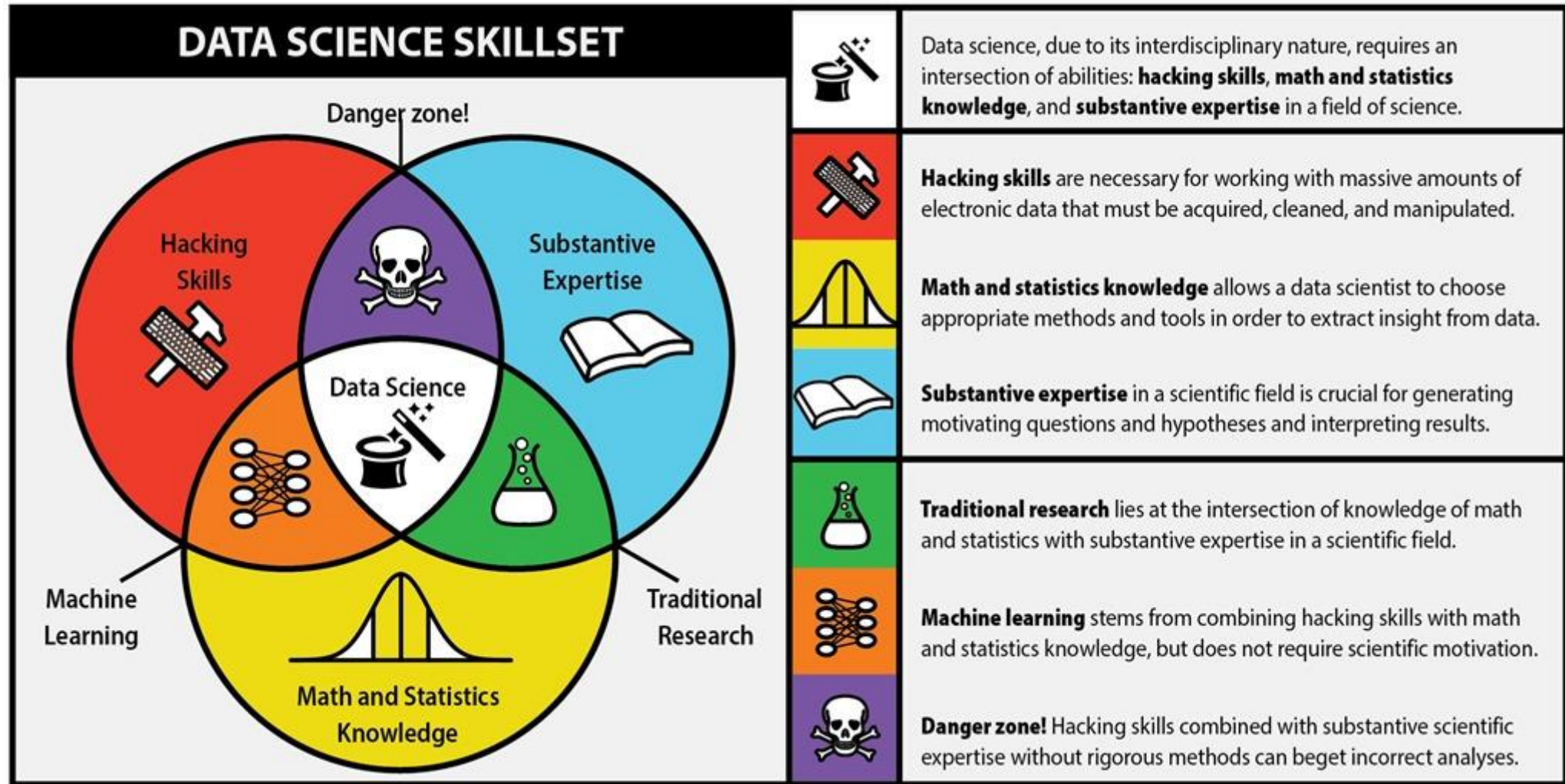
Ability to work autonomously



Adaptability

Source: Indeed

Modern Biostatistics



Modern Biostatistics

Biostatisticians are increasingly needing to combine skills in other domains

- Statistical theory and experimental designs
- Programming (R, Python, SAS, SQL)
- Clinical and biomedical knowledge (trials, epidemiology, public health)
- Regulatory knowledge (FDA guidelines, etc.)
- Machine learning and AI literacy
- Communication with medical and public-health stakeholders

Types of Data

In order to analyze the data, it is helpful to know the types of variables we are using.

Categorical

Binary/Dichotomous: only two options (*card color red vs. black*)

Nominal: no natural order (*card suit*)

Ordinal: has a natural order (*A, 2, 3, ..., 10, J, Q, K*)



Types of Data

Count/Discrete

Countable, ordered whole numbers (*# of students, # of strokes*)

Continuous

Ordered numerical data that can, in theory, take on any value (*height, weight, age, cholesterol level*)

Note:

- Continuous data can sometimes be treated as categorical.
- Discrete data can sometimes be approximated as continuous.

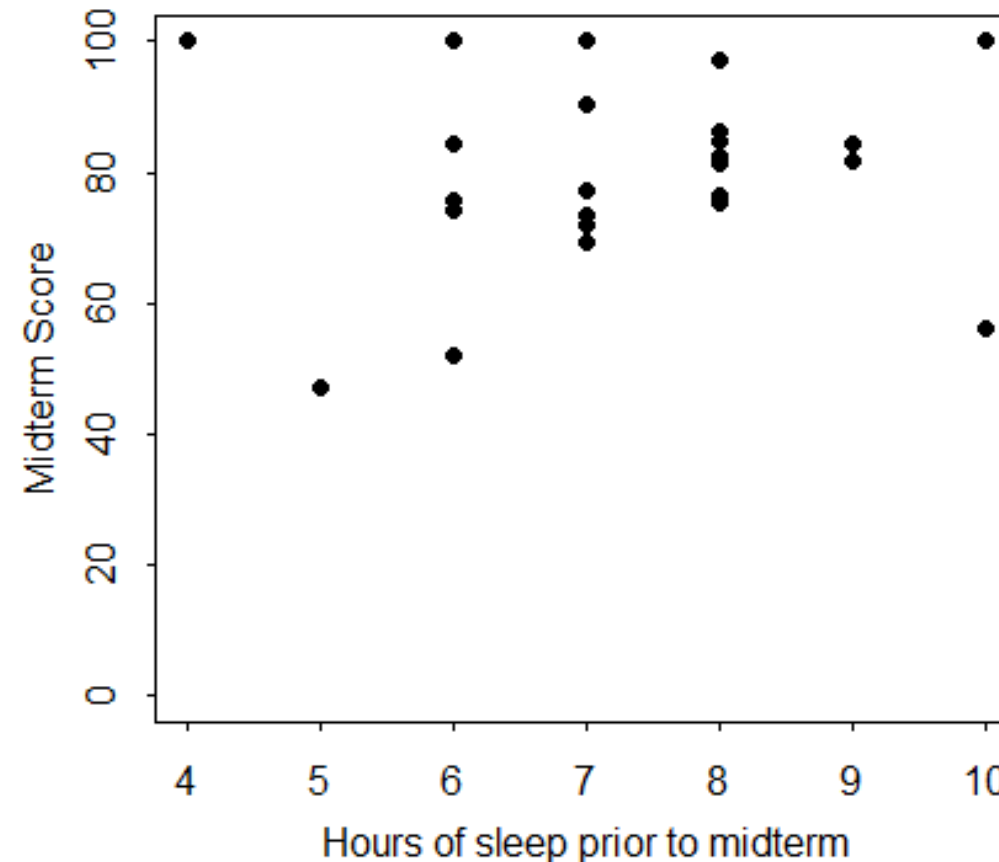
How can we begin to think statistically?

Biggest advice: don't take things that seem obvious or like "common sense" at face value!

Recognize the role that randomness and chance plays in the processes you are studying.

Quantifying Statistical Evidence

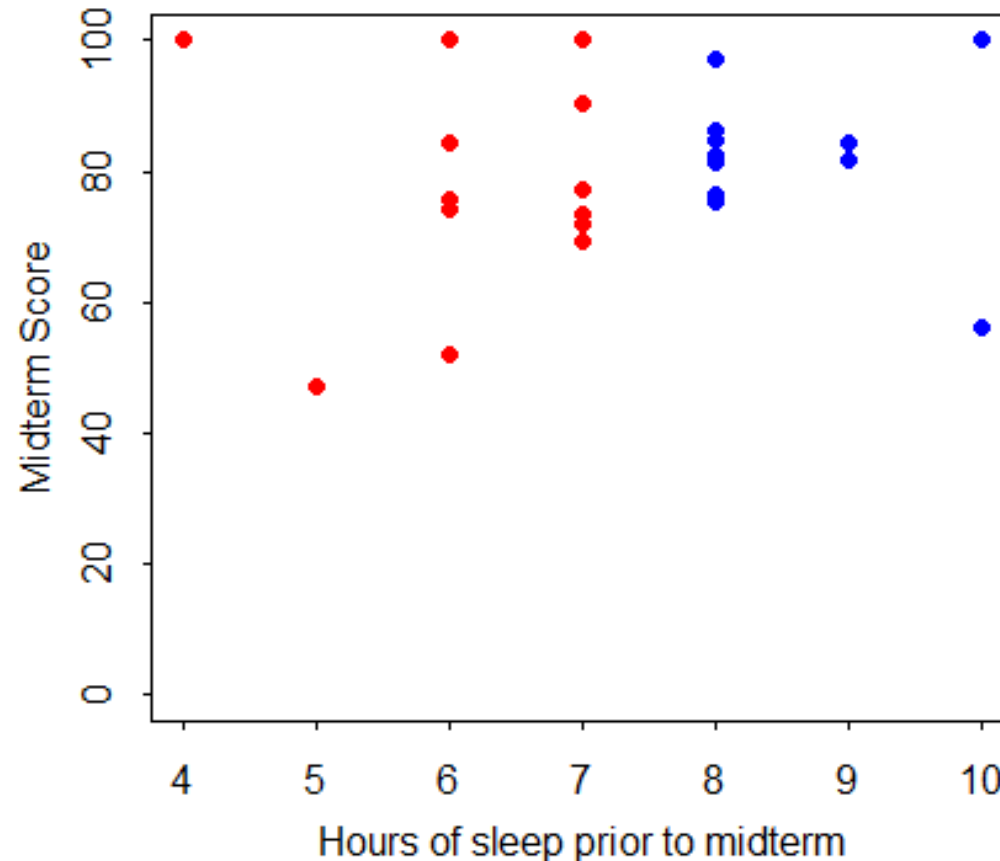
Example: Is midterm grade associated with sleep duration prior to midterm?



Is there an association?
Does it seem obvious?
Could you quantify this association?

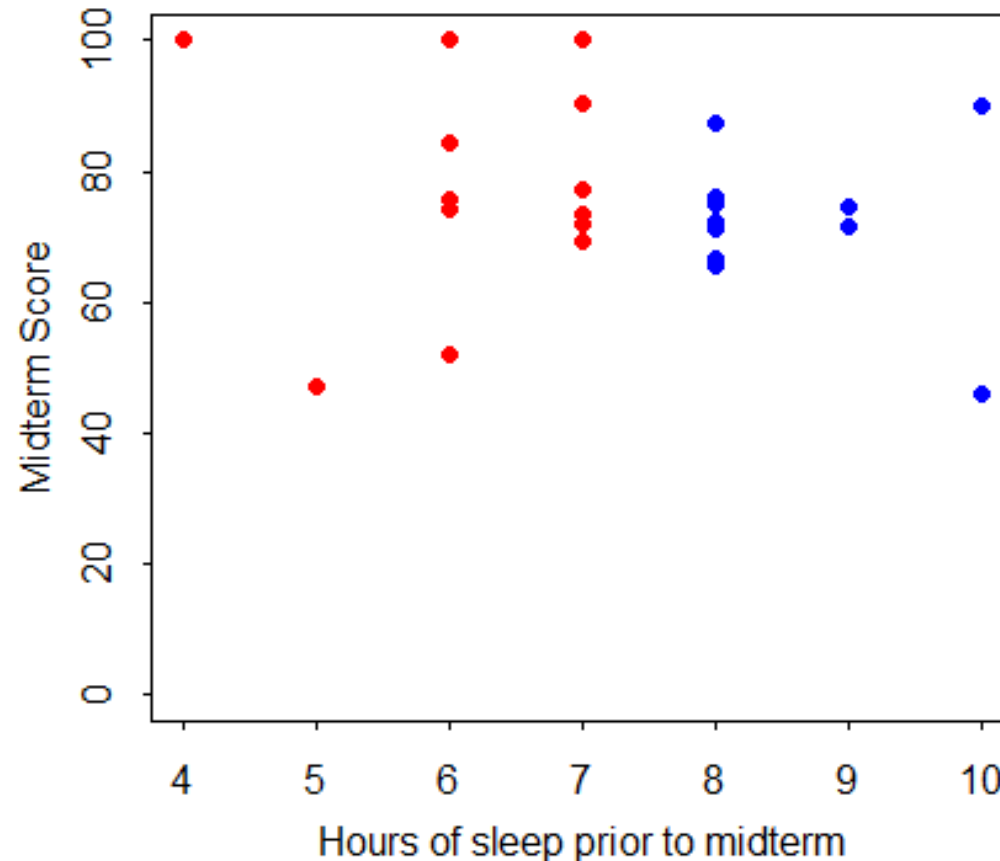
Quantifying Statistical Evidence

What if we just want to know if there's a difference in midterm score based on whether you slept a short (<8 hours) vs. long (≥ 8 hours) amount of time?



Quantifying Statistical Evidence

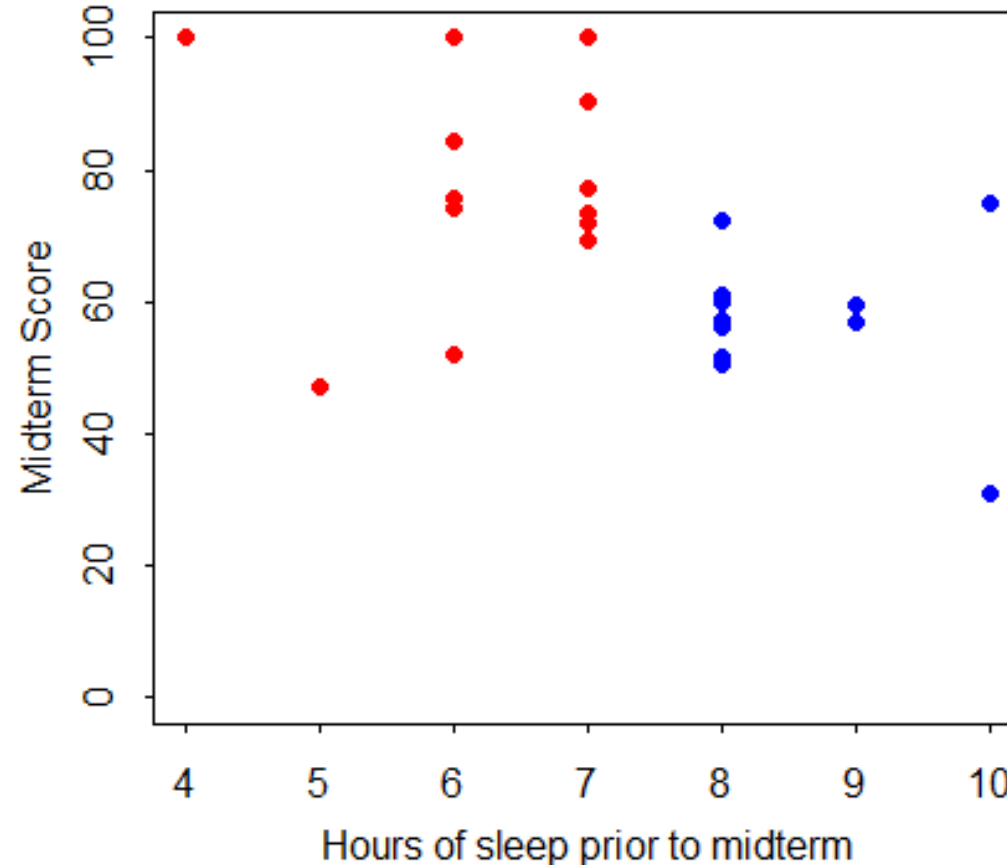
What if we just want to know if there's a difference in midterm score based on whether you slept a short (<8 hours) vs. long (≥ 8 hours) amount of time?



How about now?

Quantifying Statistical Evidence

What if we just want to know if there's a difference in midterm score based on whether you slept a short (<8 hours) vs. long (≥ 8 hours) amount of time?



How about now?

The Sample

One of the reasons we need statistics is that we aren't omniscient.

- We don't know who citizens are going to vote for in the next election.
- We don't know what everybody thought about that new restaurant.
- We don't know the vaping habits of youth in Los Angeles county.

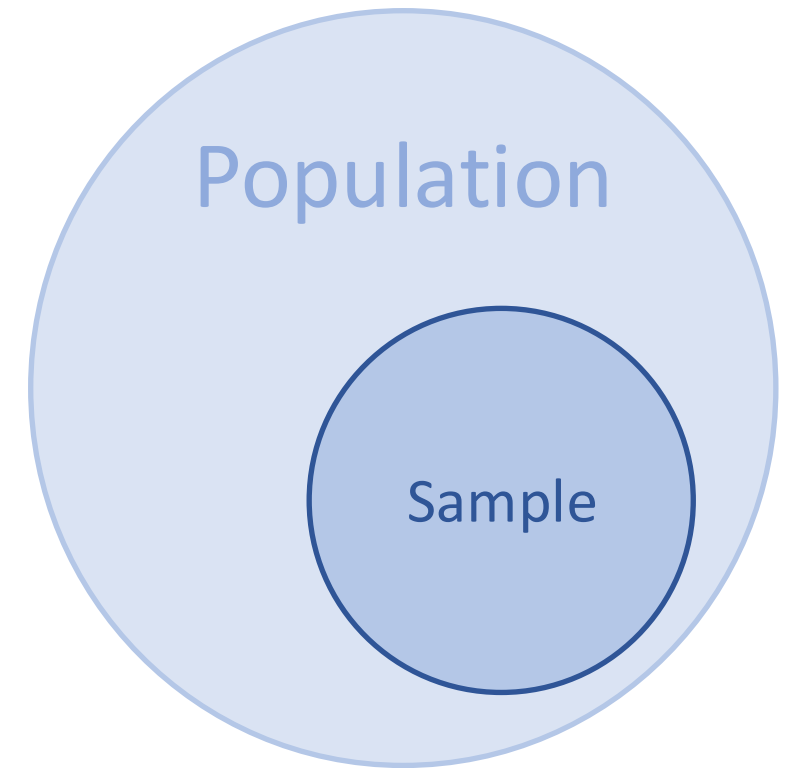
So what do we do?



Take A Sample!

We sample people/things that belong to the population of interest.

- We don't know who citizens are going to vote for in the next election.
- We don't know the vaping habits of youth in Los Angeles county.
- We don't know what everybody thought about that new restaurant.

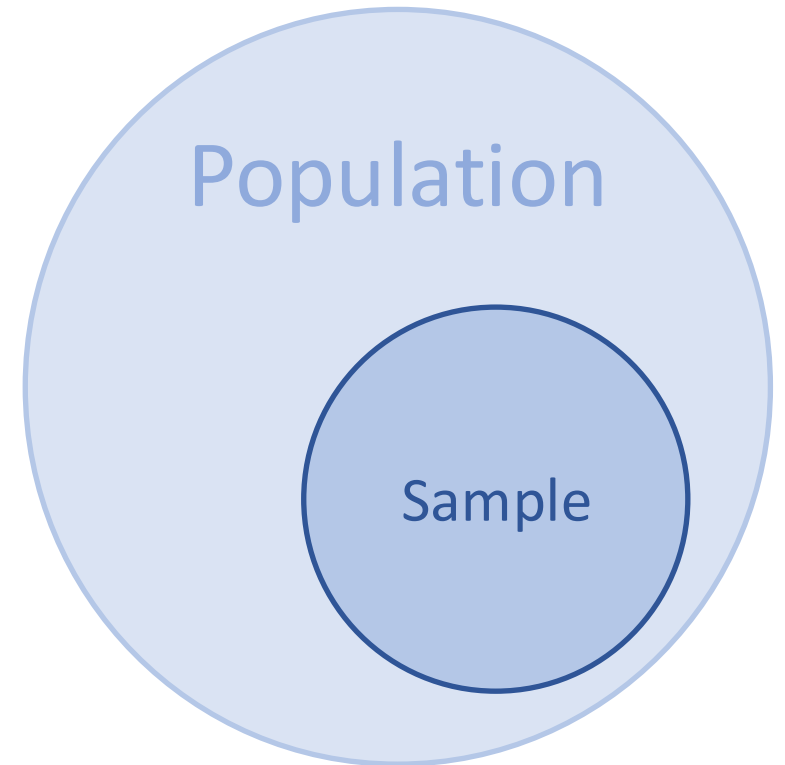


Take A Sample!

We sample people/things that belong to the population of interest.

- We don't know who citizens are going to vote for in the next election.
Ask the people in this room.
- We don't know the vaping habits of everybody in Los Angeles county.
Take a survey of your friends/family.
- We don't know what everybody thought about that new restaurant.
Check Yelp.

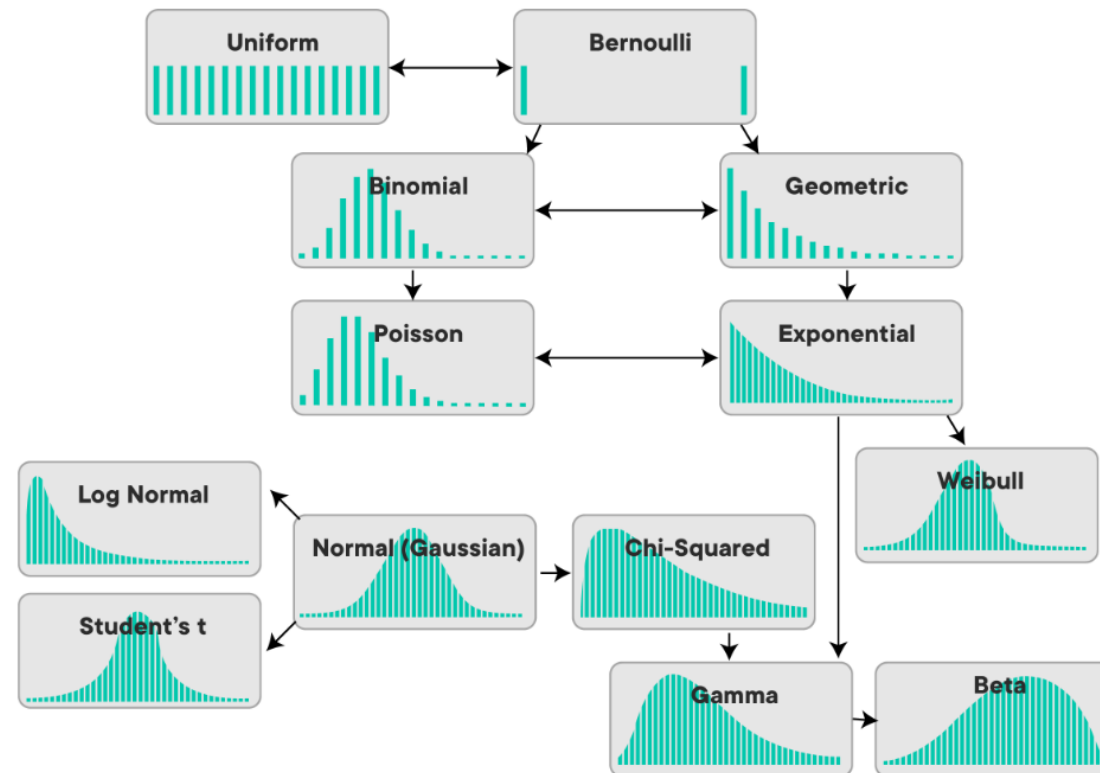
Do you think these are good ways to sample?

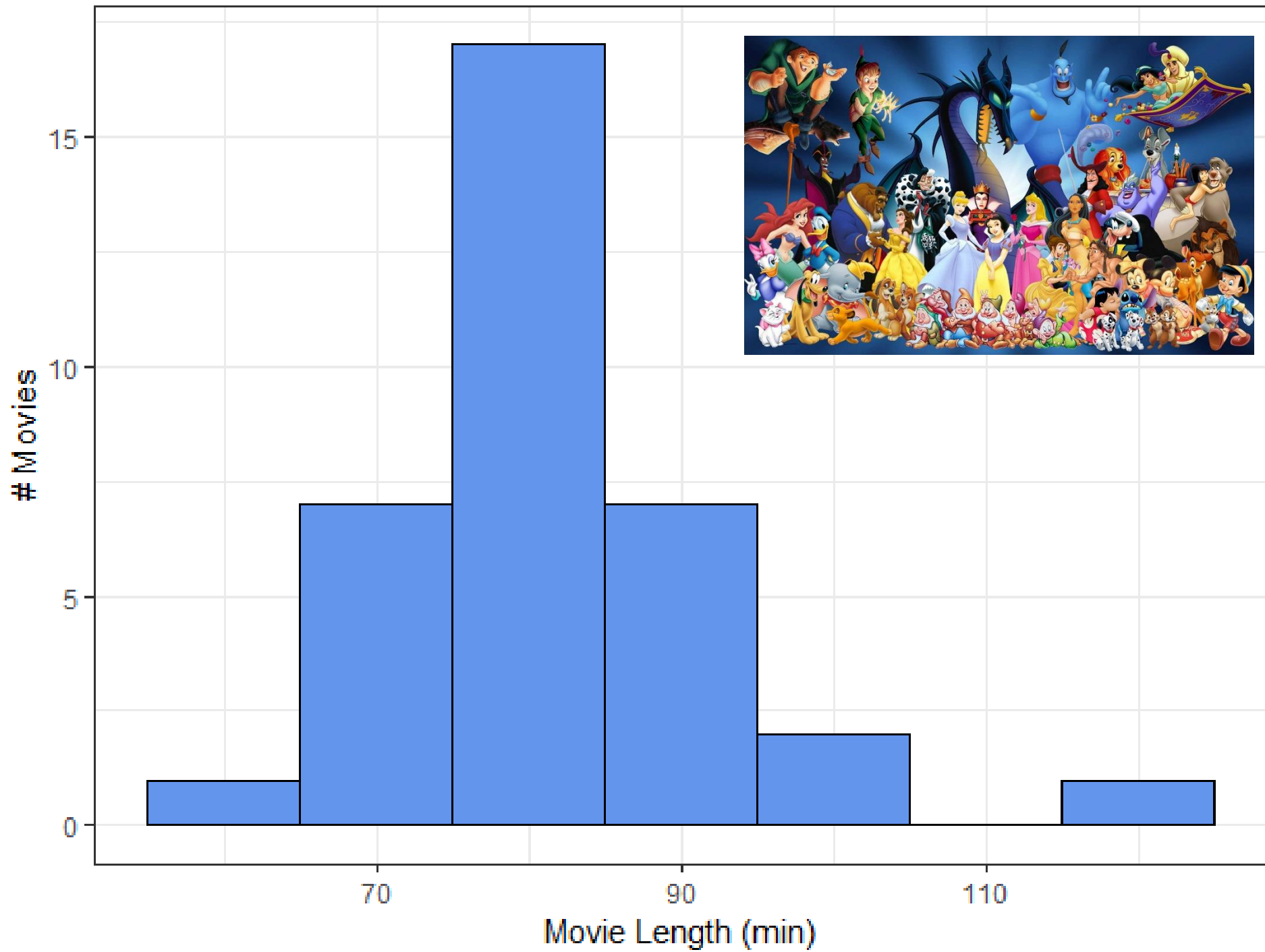


As you can see, this approach is subject to bias.

Distributions

When we sample, we get data. And when we get data, see that different variables are distributed in different ways.





Some Notation

These three quantities will be useful

Sum:

$$\sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

Add up all the values.

Sum of squares:

$$\sum_{i=1}^n x_i^2 = x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2$$

Square each value and then add them all up.

Square of the sum:

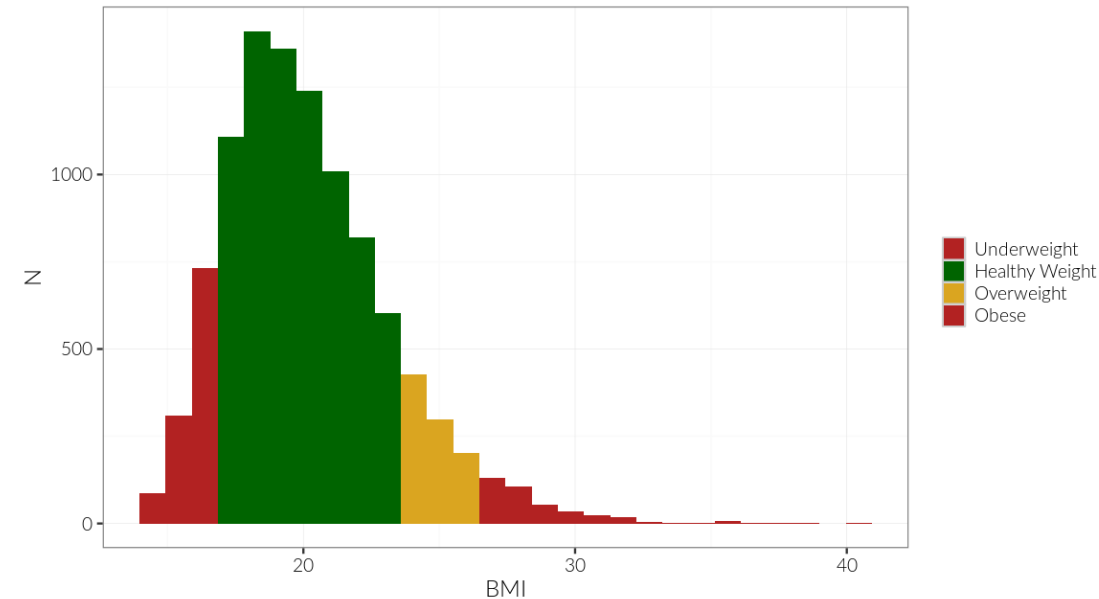
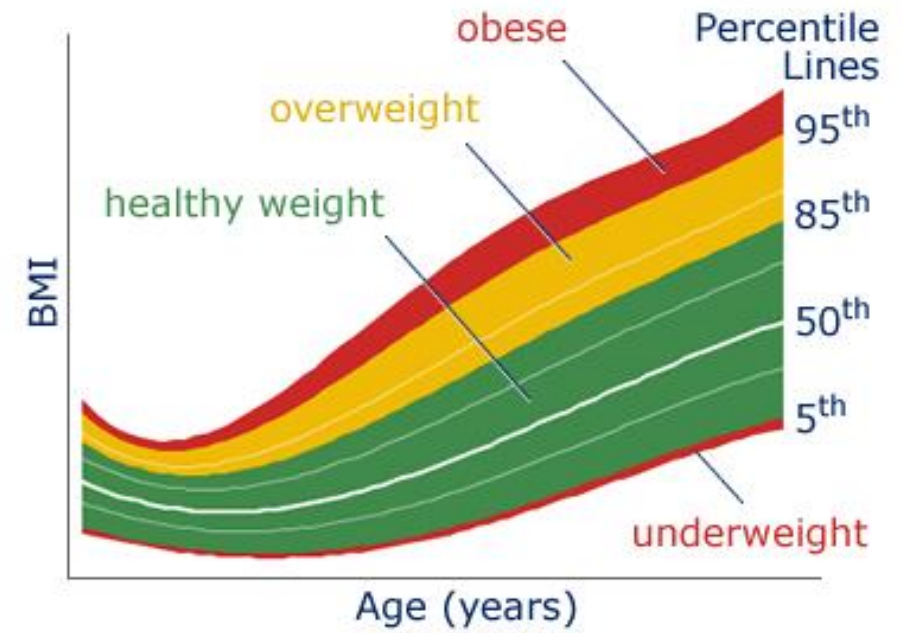
$$\left(\sum_{i=1}^n x_i \right)^2 = (x_1 + x_2 + x_3 + \cdots + x_n)^2$$

Add the values up and then square that result.

Percentiles

An X^{th} percentile is the value of which $X\%$ of the data is to the left of on the distribution.

Example: BMI in children



Central Tendency

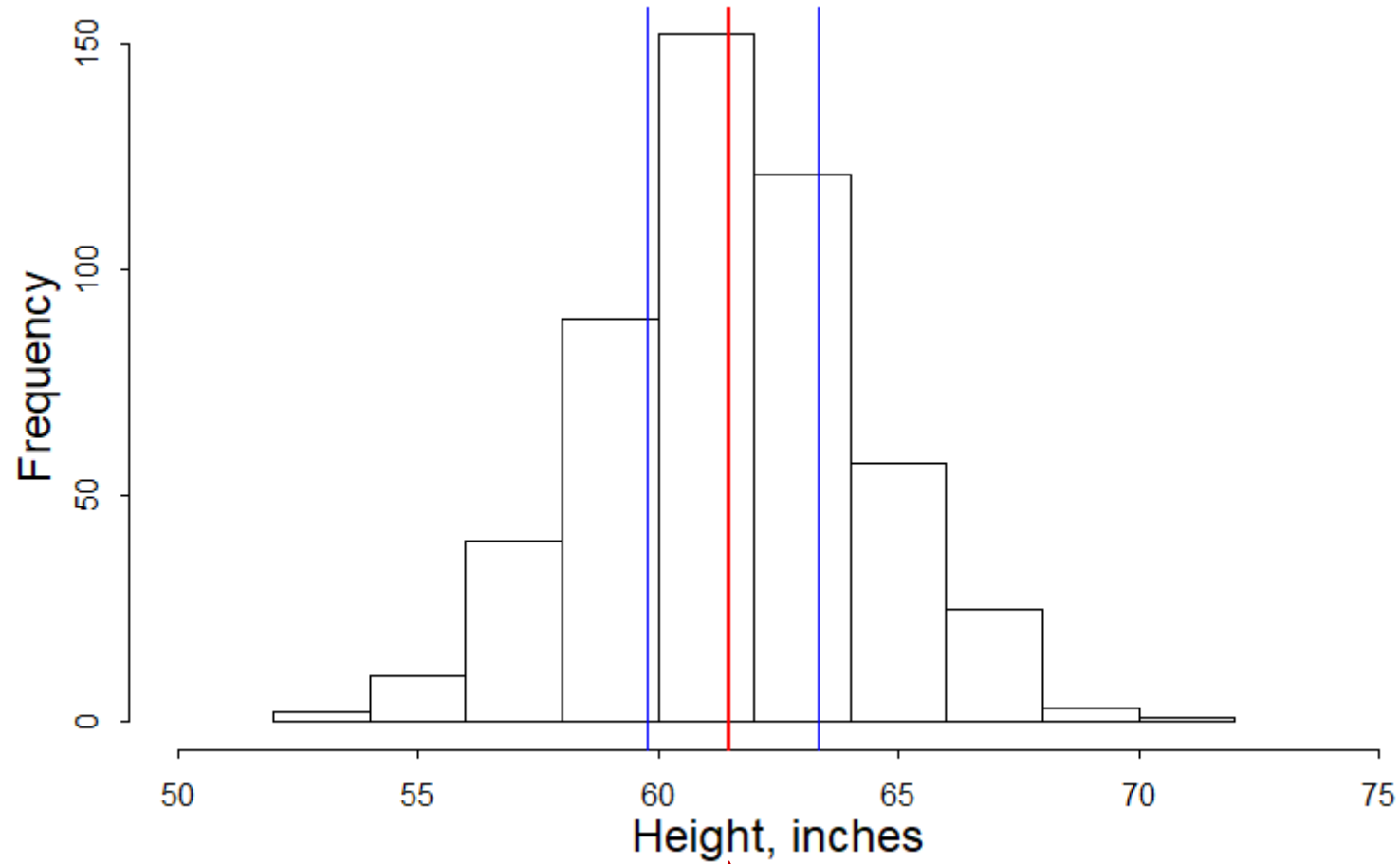
Central Tendency is a way of describing a “typical” value in the data set.

Measures of “central tendency”; “middle” of the data

- **Median:** the 50th percentile; the middle value
- **Mean (average):**
- **Mode:** value occurring most often (discrete/categorical data)
 - For continuous data, the value at the “peak”

Fun Fact: Most people have an above-average number of legs.

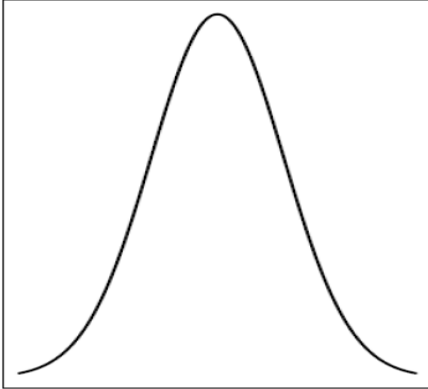
An Example with Height



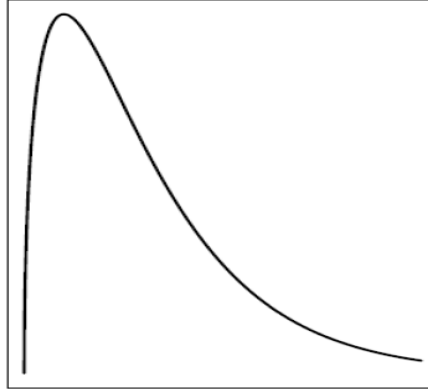
Mean $\approx$ Median $\approx$ Mode
61.5 in

A Note on Common Distribution Shapes

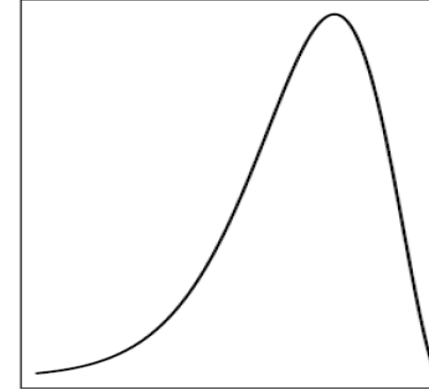
**Symmetrical
and bell shaped**



**Positively skewed
or skewed to the right**



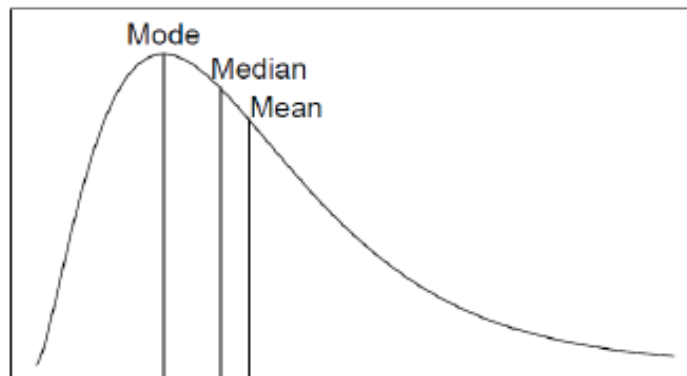
**Negatively skewed
or skewed to the left**



Positive skew/skewed to the right

- Longer tail in high values
- Mean > median > mode

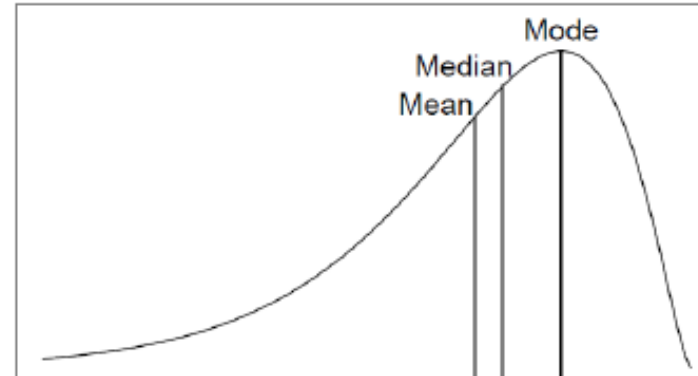
Positively skewed or skewed to the right



Negative skew/skewed to the left

- Longer tail in low values
- Mode > Median > Mean

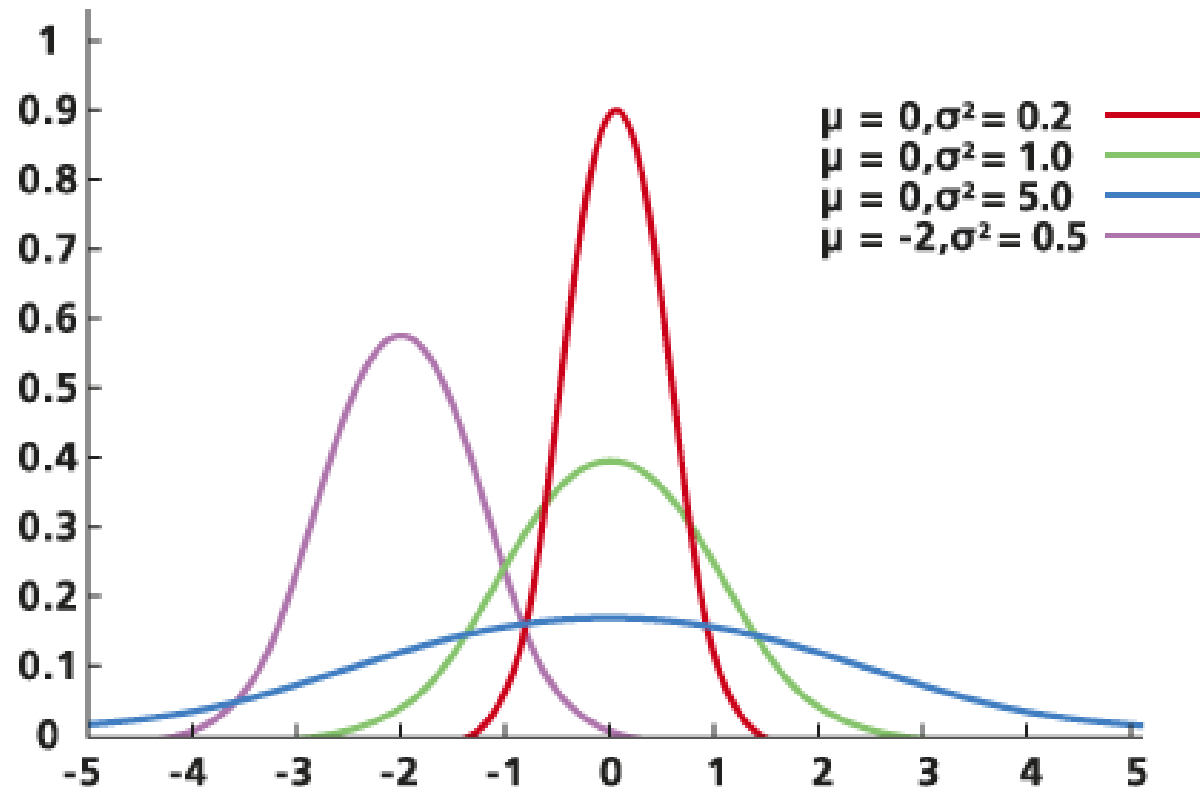
Negatively skewed or skewed to the left



What types of variables
would follow these
distributions?

Central Tendency vs. Spread

Measures of central tendency alone do not adequately describe the distribution of the data.

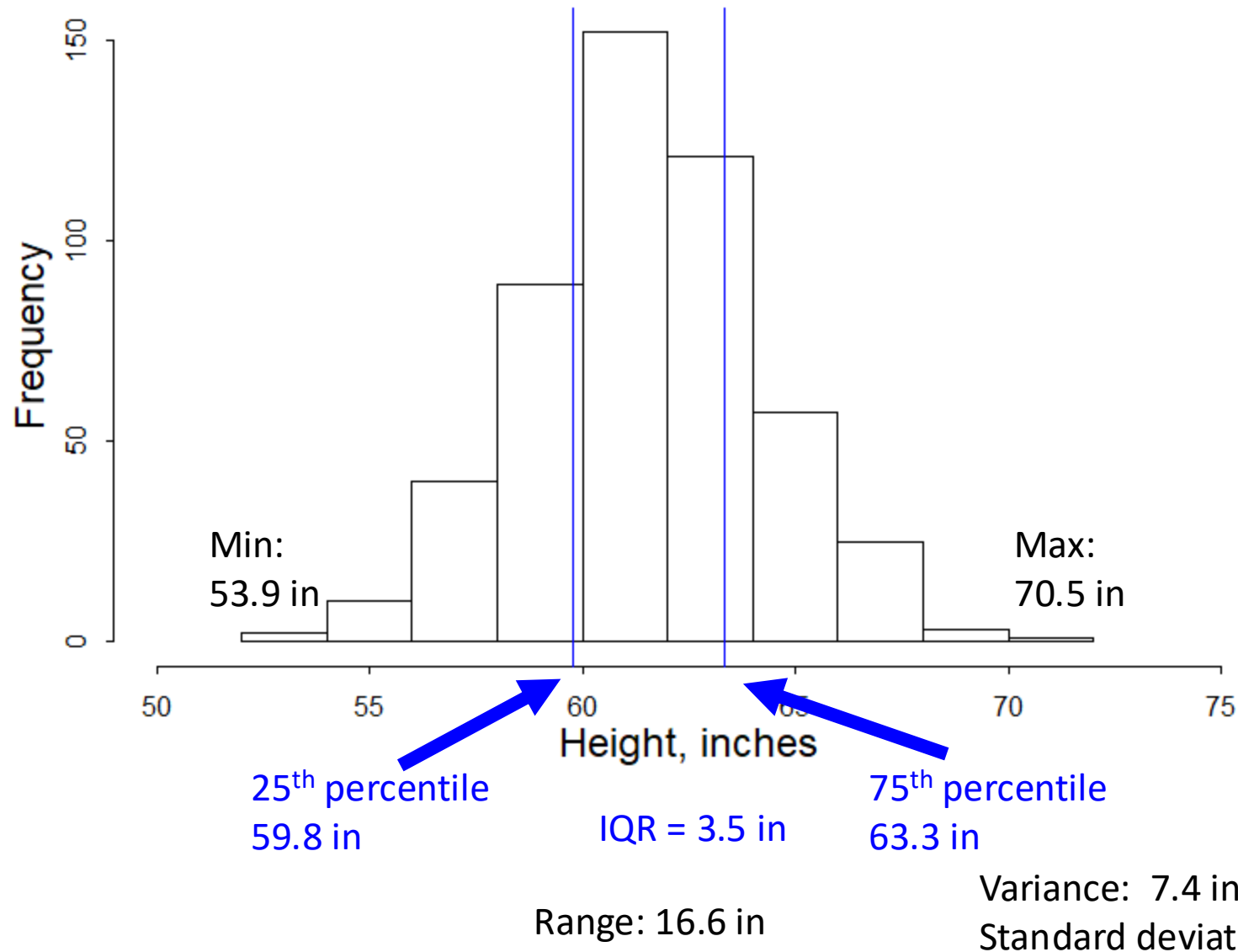


We also need to know
the spread!

Measures of Spread

- Range: difference between largest and smallest value
- Interquartile range (IQR):
difference between 75th and 25th percentile values
- Variance: “average” squared deviation from the mean
 - units: inches²
- Standard deviation (SD): square root of variance
 - unit: inches

Height: Measures of Spread

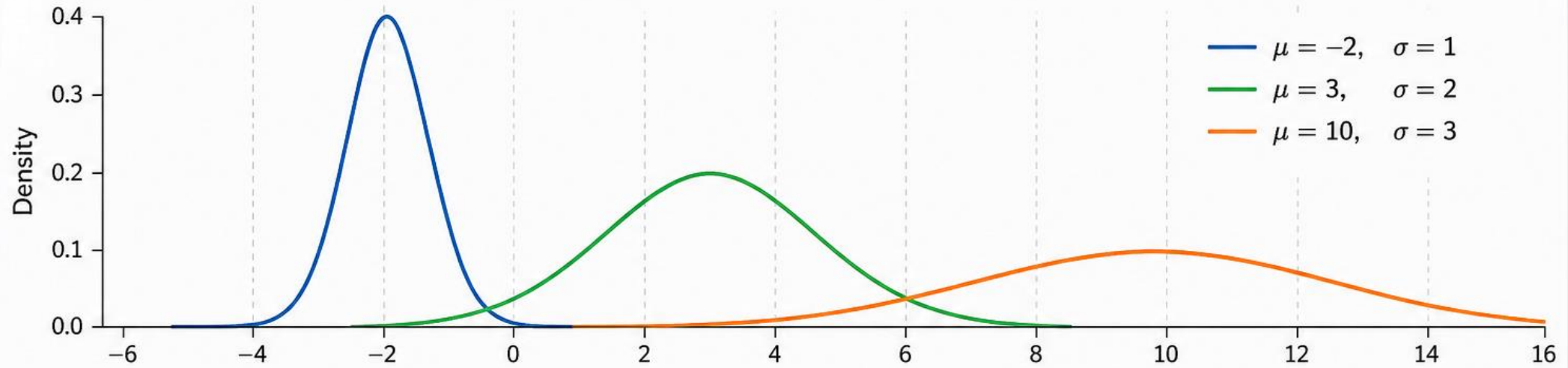


Standardization: Scaling and Centering

- You might want to compare values coming from distributions in different populations
- Standardization can help:
- “Center” by subtracting the mean $X'_i = X_i - \bar{X}$
 - New mean = 0
- “Scaled” by dividing by the standard deviation
 - New standard deviation = 1
 - Now a 1 unit change in X'' is equivalent to a SD change in X

$$X''_i = \frac{X'_i}{s} = \frac{X_i - \bar{X}}{s}$$

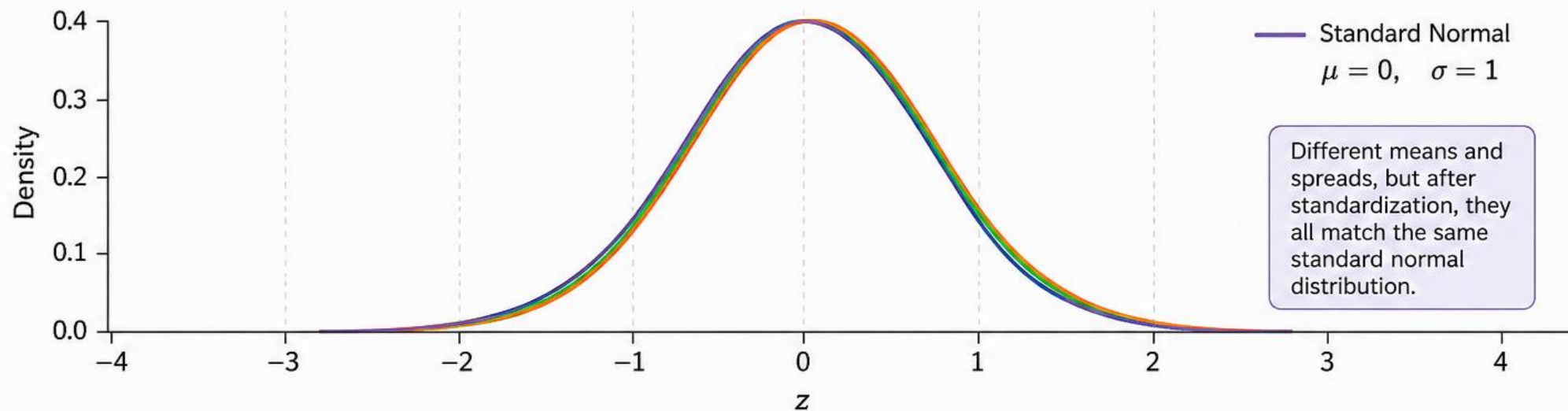
Three Different Normal Distributions



Standardization $Z = \frac{X - \mu}{\sigma}$

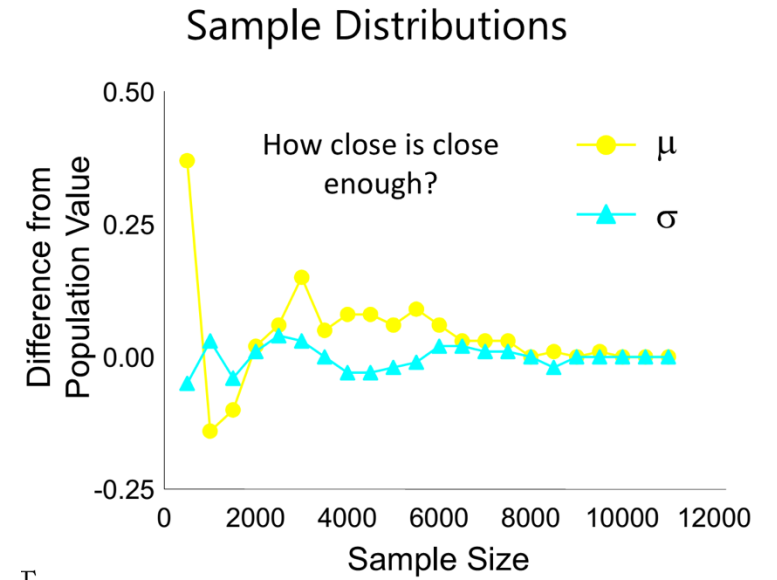


After Standardization: All Three Conform to the Standard Normal Distribution



Sample vs. Population Distribution

- Typically we never get to observe the actual population distribution
- Estimate population parameters by sampling from the population
- Ability to accurately characterize population depends on representativeness of sample and sample size
- In practice, not all data are normal
- Statistical methods fall into two broad categories
 - Parametric: assumes data follow a given distribution
 - Non-parametric: does not assume a distribution
- Some statistics are more robust than others to deviation from the assumed distribution



Recap

- “About me” - AKA - “Why the heck do I like stats?”
- What is biostatistics?
- Types of data
- Thinking statistically
- Describing a sample: distributions, percentiles, central tendency, variation